

Linear Algebra, Regression First

Companion Note: Proofs of the Results Stated in the Slides

Ignacio Urbina

PhD Math Camp

The slides for the linear-algebra unit state a number of results without proof and mark each one with a pointer to this note. This note proves them. Every result below opens with the statement exactly as it appears on the slide, so you can read the proof next to the claim. Section numbers here are the ones the slides cite; they do not match the section numbers of the deck itself.

Nothing here is examinable in Math Camp. The note exists so that no claim in the unit rests on “one can show.”

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1 Convexity and the First-Order Condition

The slide “Why the FOC Gives a Minimum” argues that $\text{SSR}(a, b)$ is convex because it is a sum of squares of linear expressions, and that a stationary point of a convex function is its global minimum. Both claims are proved here, for functions of any number of variables.

Throughout, $f : \mathbb{R}^m \rightarrow \mathbb{R}$ and points of $\mathbb{R}^m$ are written as vectors.

Definition 1.1 (Convex function). $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is *convex* if for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^m$ and all $t \in [0, 1]$,

$$f((1-t)\mathbf{u} + t\mathbf{v}) \leq (1-t)f(\mathbf{u}) + tf(\mathbf{v}).$$

Geometrically: the graph of f lies on or below the chord joining any two of its points.

Lemma 1.2 (The square is convex). $g(z) = z^2$ is convex on $\mathbb{R}$.

Proof. Fix $u, v \in \mathbb{R}$ and $t \in [0, 1]$. Then

$$\begin{aligned} (1-t)u^2 + tv^2 - ((1-t)u + tv)^2 &= (1-t)u^2 + tv^2 - (1-t)^2u^2 - 2t(1-t)uv - t^2v^2 \\ &= (1-t)(1 - (1-t))u^2 + t(1-t)v^2 - 2t(1-t)uv \\ &= t(1-t)u^2 + t(1-t)v^2 - 2t(1-t)uv \\ &= t(1-t)(u-v)^2 \geq 0, \end{aligned}$$

since $t(1-t) \geq 0$ on $[0, 1]$. Rearranging gives the definition. $\square$

Lemma 1.3 (Composition with an affine map preserves convexity). Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be convex and let $\ell : \mathbb{R}^m \rightarrow \mathbb{R}$ be affine, that is, $\ell(\mathbf{w}) = \mathbf{c} \cdot \mathbf{w} + d$ for some fixed $\mathbf{c} \in \mathbb{R}^m$ and $d \in \mathbb{R}$. Then $h(\mathbf{w}) = g(\ell(\mathbf{w}))$ is convex.

Proof. An affine map takes convex combinations to convex combinations:

$$\ell((1-t)\mathbf{u} + t\mathbf{v}) = \mathbf{c} \cdot ((1-t)\mathbf{u} + t\mathbf{v}) + d = (1-t)(\mathbf{c} \cdot \mathbf{u} + d) + t(\mathbf{c} \cdot \mathbf{v} + d) = (1-t)\ell(\mathbf{u}) + t\ell(\mathbf{v}).$$

Apply g and use its convexity:

$$h((1-t)\mathbf{u} + t\mathbf{v}) = g(\ell((1-t)\mathbf{u} + t\mathbf{v})) \leq (1-t)g(\ell(\mathbf{u})) + tg(\ell(\mathbf{v})) = (1-t)h(\mathbf{u}) + th(\mathbf{v}). \quad \square$$

Lemma 1.4 (Sums of convex functions are convex). If $h_1, \dots, h_n : \mathbb{R}^m \rightarrow \mathbb{R}$ are convex, so is $h_1 + \dots + h_n$.

Proof. Add the n inequalities of Definition 1.1, one for each h_i , at the same $\mathbf{u}, \mathbf{v}, t$. $\square$

Proposition 1.5 (SSR is convex). Slide statement. Each term $e_i^2 = (y_i - a - bx_i)^2$ is the square of an expression linear in (a, b) ; the square is convex; a sum of convex functions is convex. Hence $\text{SSR}(a, b)$ is convex.

More generally, for any $n \times p$ matrix X and $\mathbf{y} \in \mathbb{R}^n$, the function $\text{SSR}(\boldsymbol{\beta}) = \|\mathbf{y} - X\boldsymbol{\beta}\|^2$ is convex on $\mathbb{R}^p$.

Proof. Write $\text{SSR}(\boldsymbol{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i \cdot \boldsymbol{\beta})^2$, where $\mathbf{x}_i'$ is row i of X . The map $\boldsymbol{\beta} \mapsto y_i - \mathbf{x}_i \cdot \boldsymbol{\beta}$ is affine in $\boldsymbol{\beta}$ (take $\mathbf{c} = -\mathbf{x}_i$, $d = y_i$ in Lemma 1.3). Composing with the square (Lemma 1.2) gives a convex function of $\boldsymbol{\beta}$ for each i . Summing over i (Lemma 1.4) gives convexity of SSR. The simple-regression case is $p = 2$, $\mathbf{x}_i = (1, x_i)$, $\boldsymbol{\beta} = (a, b)$. $\square$

Theorem 1.6 (Stationary points of convex functions are global minima). Slide statement. A bowl has one bottom: the point where both partial derivatives vanish is the global minimum.

Precisely: let $f : \mathbb{R}^m \rightarrow \mathbb{R}$ be convex with continuous partial derivatives, and suppose $\nabla f(\mathbf{u}^*) = \mathbf{0}$, i.e. every partial derivative of f is zero at $\mathbf{u}^*$. Then $f(\mathbf{u}^*) \leq f(\mathbf{v})$ for every $\mathbf{v} \in \mathbb{R}^m$.

Proof. Fix any $\mathbf{v}$ and let $\mathbf{d} = \mathbf{v} - \mathbf{u}^*$. Define the one-variable function $\varphi(t) = f(\mathbf{u}^* + t\mathbf{d})$ for $t \in [0, 1]$: the values of f along the segment from $\mathbf{u}^*$ to $\mathbf{v}$.

Step 1: $\varphi'(0) = 0$. By the chain rule for a function of several variables,

$$\varphi'(t) = \sum_{j=1}^m \frac{\partial f}{\partial u_j}(\mathbf{u}^* + t\mathbf{d}) d_j = \nabla f(\mathbf{u}^* + t\mathbf{d}) \cdot \mathbf{d}.$$

At $t = 0$ this is $\nabla f(\mathbf{u}^*) \cdot \mathbf{d} = \mathbf{0} \cdot \mathbf{d} = 0$.

Step 2: *convexity bounds the difference quotient.* Convexity of f with $\mathbf{u} = \mathbf{u}^*$ gives, for $t \in (0, 1]$,

$$f(\mathbf{u}^* + t\mathbf{d}) = f((1-t)\mathbf{u}^* + t\mathbf{v}) \leq (1-t)f(\mathbf{u}^*) + tf(\mathbf{v}),$$

so

$$\frac{f(\mathbf{u}^* + t\mathbf{d}) - f(\mathbf{u}^*)}{t} \leq f(\mathbf{v}) - f(\mathbf{u}^*).$$

The left side is $(\varphi(t) - \varphi(0))/t$.

Step 3: let $t \rightarrow 0^+$. The left side tends to $\varphi'(0) = 0$, and the right side does not depend on t . Hence $0 \leq f(\mathbf{v}) - f(\mathbf{u}^*)$, which is the claim. $\square$

Remark 1.7. The theorem needs no second-order condition. That is why the slides can skip the Hessian: for a convex objective, the FOC alone certifies a global minimum. The price is that convexity must be established separately, which Proposition 1.5 does for SSR.

2 The Normal Equations in Full

The slides “Taking the Derivatives,” “The Normal Equations,” and “Solving in General” skip algebra. Here is every step.

Data: $(x_1, y_1), \dots, (x_n, y_n)$. Define $e_i = y_i - a - bx_i$ and $\text{SSR}(a, b) = \sum_{i=1}^n e_i^2$. Write $\bar{x} = \frac{1}{n} \sum x_i$, $\bar{y} = \frac{1}{n} \sum y_i$.

Proposition 2.1 (The partial derivatives).

$$\frac{\partial \text{SSR}}{\partial a} = -2 \sum_{i=1}^n e_i, \quad \frac{\partial \text{SSR}}{\partial b} = -2 \sum_{i=1}^n x_i e_i.$$

Proof. Each term $e_i^2 = (y_i - a - bx_i)^2$ is differentiated by the chain rule: outer derivative $2(y_i - a - bx_i) = 2e_i$, inner derivative $\partial(y_i - a - bx_i)/\partial a = -1$ and $\partial(y_i - a - bx_i)/\partial b = -x_i$. The derivative of a sum is the sum of the derivatives. $\square$

Proposition 2.2 (The normal equations). Slide statement. *The FOC $\partial \text{SSR}/\partial a = \partial \text{SSR}/\partial b = 0$ is equivalent to*

$$\sum_i y_i = na + b \sum_i x_i, \quad \sum_i x_i y_i = a \sum_i x_i + b \sum_i x_i^2.$$

Proof. Setting the two partials of Proposition 2.1 to zero and dividing by -2 :

$$\sum_i (y_i - a - bx_i) = 0, \quad \sum_i x_i (y_i - a - bx_i) = 0.$$

Distribute the sums. In the first, $\sum_i a = na$ and $\sum_i bx_i = b \sum_i x_i$, so $\sum_i y_i - na - b \sum_i x_i = 0$. In the second, $\sum_i x_i y_i - a \sum_i x_i - b \sum_i x_i^2 = 0$. Rearrange. $\square$

Proposition 2.3 (Closed form). Slide statement.

$$b = \frac{\sum_i x_i y_i - n \bar{x} \bar{y}}{\sum_i x_i^2 - n \bar{x}^2} = \frac{S_{xy}}{S_{xx}}, \quad a = \bar{y} - b \bar{x},$$

provided $S_{xx} = \sum_i (x_i - \bar{x})^2 \neq 0$, i.e. the x_i are not all equal.

Proof. Divide the first normal equation by n : $\bar{y} = a + b \bar{x}$, so $a = \bar{y} - b \bar{x}$. Substitute into the second:

$$\begin{aligned} \sum_i x_i y_i &= (\bar{y} - b \bar{x}) \sum_i x_i + b \sum_i x_i^2 = (\bar{y} - b \bar{x}) n \bar{x} + b \sum_i x_i^2 \\ &= n \bar{x} \bar{y} - b n \bar{x}^2 + b \sum_i x_i^2. \end{aligned}$$

Collect the terms in b :

$$b \left(\sum_i x_i^2 - n \bar{x}^2 \right) = \sum_i x_i y_i - n \bar{x} \bar{y}.$$

Finally, the two identities

$$\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - n \bar{x}^2, \quad \sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i x_i y_i - n \bar{x} \bar{y}$$

follow by expanding the left sides and using $\sum_i x_i = n \bar{x}$, $\sum_i y_i = n \bar{y}$:

$$\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i x_i y_i - \bar{y} \sum_i x_i - \bar{x} \sum_i y_i + n \bar{x} \bar{y} = \sum_i x_i y_i - n \bar{x} \bar{y},$$

and the first is the case $y = x$. The divisor S_{xx} is a sum of squares, zero only if every $x_i = \bar{x}$. $\square$

Corollary 2.4 (Two identities the fitted line satisfies). *At the least-squares* (a, b) , $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$.

Proof. These are the two equations set to zero in the proof of Proposition 2.2. $\square$

Example 2.5 (The ten states). Here x_i is state i 's unemployment rate relative to the national rate and y_i its incumbent vote share:

$$\mathbf{x} = (-2, -1, -1, 0, 1, 1, 2, 3, 3, 4), \quad \mathbf{y} = (55, 55, 53, 53, 52, 48, 49, 45, 47, 43).$$

So $n = 10$, $\sum x_i = 10$, $\sum y_i = 500$, $\sum x_i^2 = 46$, $\sum x_i y_i = 428$, giving $\bar{x} = 1$, $\bar{y} = 50$, $S_{xx} = 46 - 10 \cdot 1 = 36$, $S_{xy} = 428 - 10 \cdot 1 \cdot 50 = -72$, $b = -72/36 = -2$, $a = 50 - (-2)(1) = 52$. The residuals are $(-1, 1, -1, 1, 2, -2, 1, -1, 1, -1)$; their sum is 0 and $\sum x_i e_i = 2 - 1 + 1 + 0 + 2 - 2 + 2 - 3 + 3 - 4 = 0$, confirming Corollary 2.4. SSR = 16.

Remark 2.6 (Why x is centered on the national rate). Shifting a predictor by a constant, $x_i \mapsto x_i - c$, leaves S_{xx} , S_{xy} , the slope b , every residual, SSR and the correlation unchanged; only the intercept moves, to $a + bc$. The slides use $c = 5$ for a reason that is about the *picture*, not the fit. Drawn in the (b, a) plane, the two normal equations are lines of slope $-\bar{x}$ and $-\sum x_i^2 / \sum x_i$, whose ratio is

$$\frac{n \sum x_i^2}{(\sum x_i)^2} = 1 + \frac{S_{xx}}{n \bar{x}^2} = 1 + \text{CV}^2(x).$$

For raw unemployment rates this is 1.10, so the lines are within 2.7° of each other at the best possible plot aspect and the picture of two lines crossing is unreadable. Centering raises the ratio to 4.6 and the separation to 40° .

3 Row Operations Preserve the Solution Set

Proposition 3.1 (Row operations preserve the solution set). Slide statement. *If $[A' \mid \mathbf{b}']$ is obtained from $[A \mid \mathbf{b}]$ by row operations, then $A'\mathbf{x} = \mathbf{b}'$ and $A\mathbf{x} = \mathbf{b}$ have exactly the same solutions.*

Proof. It is enough to prove it for a single row operation; a sequence of them is handled by applying the single case repeatedly. Write the equations of the system as $E_1, \dots, E_m$, where E_i is $a_{i1}x_1 + \dots + a_{in}x_n = b_i$. A vector $\mathbf{x}$ solves the system if and only if it satisfies every E_i .

Swap. Swapping E_i and E_j changes the order in which the equations are listed, not which equations there are. The solution set is unchanged.

Scale. Replace E_i by cE_i with $c \neq 0$: the equation $c(a_{i1}x_1 + \dots + a_{in}x_n) = cb_i$. If $\mathbf{x}$ satisfies E_i , multiply both sides by c : it satisfies cE_i . If $\mathbf{x}$ satisfies cE_i , multiply both sides by $1/c$ (this is where $c \neq 0$ is used): it satisfies E_i . The other equations are untouched. Same solution set.

Add. Replace E_i by $E_i + cE_j$ for some $j \neq i$, leaving E_j in place. If $\mathbf{x}$ satisfies E_i and E_j , then adding c times the second to the first shows it satisfies $E_i + cE_j$. Conversely, if $\mathbf{x}$ satisfies $E_i + cE_j$ and E_j , subtracting c times E_j from the first shows it satisfies E_i . Since E_j is still in the system in both directions, the solution sets coincide. $\square$

Remark 3.2. The proof shows why the operations are exactly these three and why the scale factor must be nonzero: each operation must be *reversible* by another row operation. Scaling by 0 is not.

4 $A\mathbf{x}$ Is Linear; Matrix Multiplication Is Composition

For A an $m \times n$ matrix and $\mathbf{x} \in \mathbb{R}^n$, $(A\mathbf{x})_i = \sum_{j=1}^n a_{ij}x_j$.

Proposition 4.1 ($\mathbf{x} \mapsto A\mathbf{x}$ is linear). Slide statement. *$F(\mathbf{x}) = A\mathbf{x}$ satisfies $F(\mathbf{u} + \mathbf{v}) = F(\mathbf{u}) + F(\mathbf{v})$ and $F(c\mathbf{u}) = cF(\mathbf{u})$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, $c \in \mathbb{R}$.*

Proof. Compare entry i on both sides.

$$(A(\mathbf{u} + \mathbf{v}))_i = \sum_j a_{ij}(u_j + v_j) = \sum_j a_{ij}u_j + \sum_j a_{ij}v_j = (A\mathbf{u})_i + (A\mathbf{v})_i,$$

$$(A(c\mathbf{u}))_i = \sum_j a_{ij}(cu_j) = c \sum_j a_{ij}u_j = c(A\mathbf{u})_i.$$

Both use only that multiplication distributes over addition in $\mathbb{R}$. $\square$

Proposition 4.2 (Matrix multiplication is composition). Slide statement. *For A $m \times n$ and B $n \times p$, define $(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$ (row i of A dotted with column j of B). Then $(AB)\mathbf{x} = A(B\mathbf{x})$ for every $\mathbf{x} \in \mathbb{R}^p$. That is, AB is the matrix of the composite transformation “first B , then A .”*

Proof. Entry i of $A(B\mathbf{x})$, expanding the inner product first:

$$(A(B\mathbf{x}))_i = \sum_{k=1}^n a_{ik}(B\mathbf{x})_k = \sum_{k=1}^n a_{ik} \sum_{j=1}^p b_{kj}x_j = \sum_{j=1}^p \left(\sum_{k=1}^n a_{ik}b_{kj} \right) x_j = \sum_{j=1}^p (AB)_{ij} x_j = ((AB)\mathbf{x})_i.$$

The middle step swaps the order of two finite sums, which is always allowed. $\square$

Corollary 4.3 (Associativity). Slide statement. $(AB)C = A(BC)$ whenever the products are defined.

Proof. Apply both sides to an arbitrary vector $\mathbf{x}$ and use Proposition 4.2 three times: $((AB)C)\mathbf{x} = (AB)(C\mathbf{x}) = A(B(C\mathbf{x})) = A((BC)\mathbf{x}) = (A(BC))\mathbf{x}$. Two matrices that agree on every vector agree on the standard basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_p$, and $M\mathbf{e}_j$ is column j of M ; so they have the same columns and are equal. $\square$

Remark 4.4 (No commutativity). The slide's S (stretch) and R (rotate) give $RS = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$ and $SR = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}$. Nothing in the definition forces $AB = BA$, and in general it fails.

5 The Cosine Formula and Cauchy–Schwarz

Proposition 5.1 (Dot product and angle in the plane). Slide statement. For nonzero $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ with angle θ between them, $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$.

Proof. Let $\theta_{\mathbf{u}}$ and $\theta_{\mathbf{v}}$ be the angles each vector makes with the positive x -axis, so that $\theta = \theta_{\mathbf{v}} - \theta_{\mathbf{u}}$ (or its negative; cosine is even, so the sign does not matter). From the definitions of sine and cosine on a right triangle with hypotenuse $\|\mathbf{u}\|$,

$$u_1 = \|\mathbf{u}\| \cos \theta_{\mathbf{u}}, \quad u_2 = \|\mathbf{u}\| \sin \theta_{\mathbf{u}}, \quad v_1 = \|\mathbf{v}\| \cos \theta_{\mathbf{v}}, \quad v_2 = \|\mathbf{v}\| \sin \theta_{\mathbf{v}}.$$

Then, by the angle-difference identity $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$,

$$\begin{aligned} \mathbf{u} \cdot \mathbf{v} &= u_1 v_1 + u_2 v_2 = \|\mathbf{u}\| \|\mathbf{v}\| (\cos \theta_{\mathbf{u}} \cos \theta_{\mathbf{v}} + \sin \theta_{\mathbf{u}} \sin \theta_{\mathbf{v}}) \\ &= \|\mathbf{u}\| \|\mathbf{v}\| \cos(\theta_{\mathbf{v}} - \theta_{\mathbf{u}}) = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta. \end{aligned} \quad \square$$

In $\mathbb{R}^n$ with $n > 3$ there is no picture to take an angle from. The slide says the formula *is* the definition of the angle there, and that it always returns a number in $[-1, 1]$. The second claim is the following theorem.

Theorem 5.2 (Cauchy–Schwarz). For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$, with equality if and only if one vector is a scalar multiple of the other.

Proof. If $\mathbf{v} = \mathbf{0}$ both sides are 0. Otherwise, for every real t ,

$$0 \leq \|\mathbf{u} - t\mathbf{v}\|^2 = (\mathbf{u} - t\mathbf{v}) \cdot (\mathbf{u} - t\mathbf{v}) = \|\mathbf{u}\|^2 - 2t(\mathbf{u} \cdot \mathbf{v}) + t^2\|\mathbf{v}\|^2.$$

This is a quadratic in t that is never negative. Choose the t that minimizes it, $t^* = (\mathbf{u} \cdot \mathbf{v})/\|\mathbf{v}\|^2$ (set the derivative in t to zero), and substitute:

$$0 \leq \|\mathbf{u}\|^2 - 2 \frac{(\mathbf{u} \cdot \mathbf{v})^2}{\|\mathbf{v}\|^2} + \frac{(\mathbf{u} \cdot \mathbf{v})^2}{\|\mathbf{v}\|^2} = \|\mathbf{u}\|^2 - \frac{(\mathbf{u} \cdot \mathbf{v})^2}{\|\mathbf{v}\|^2}.$$

Multiply through by $\|\mathbf{v}\|^2 > 0$: $(\mathbf{u} \cdot \mathbf{v})^2 \leq \|\mathbf{u}\|^2 \|\mathbf{v}\|^2$. Take square roots. Equality holds exactly when $\|\mathbf{u} - t^*\mathbf{v}\| = 0$, i.e. $\mathbf{u} = t^*\mathbf{v}$. $\square$

Definition 5.3 (Angle in $\mathbb{R}^n$). For nonzero $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the angle between them is the unique $\theta \in [0, \pi]$ with $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$. Theorem 5.2 guarantees the right side lies in $[-1, 1]$, so such a θ exists.

Proposition 5.4 (Correlation lies in $[-1, 1]$). Slide statement. $r_{xy} = \frac{\mathbf{x}_c \cdot \mathbf{y}_c}{\|\mathbf{x}_c\| \|\mathbf{y}_c\|} = \cos \theta$, so $-1 \leq r_{xy} \leq 1$, with $|r_{xy}| = 1$ exactly when the centered vectors are proportional, i.e. the data lie exactly on a line.

Proof. That the usual formula equals the vector expression is a matter of reading it: numerator $\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \mathbf{x}_c \cdot \mathbf{y}_c$, and each factor under the root is $\|\mathbf{x}_c\|^2$ or $\|\mathbf{y}_c\|^2$. The bounds and the equality case are Theorem 5.2 applied to $\mathbf{x}_c, \mathbf{y}_c$. $\square$

6 Facts About the Inverse

Throughout, A is $n \times n$ and I is the $n \times n$ identity.

Proposition 6.1 (Uniqueness). Slide statement. If A^{-1} exists, it is unique.

Proof. Suppose B and C both satisfy $AB = BA = I$ and $AC = CA = I$. Then

$$B = BI = B(AC) = (BA)C = IC = C,$$

using associativity (Corollary 4.3) in the middle. $\square$

Proposition 6.2 (Inverse of a product). Slide statement. If A and B are invertible, so is AB , and $(AB)^{-1} = B^{-1}A^{-1}$.

Proof. Check that $B^{-1}A^{-1}$ does the job from both sides:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I,$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I.$$

By uniqueness, this is the inverse. $\square$

Proposition 6.3 (Inverse of a transpose). If A is invertible, so is A' , and $(A')^{-1} = (A^{-1})'$.

Proof. Transpose $AA^{-1} = I$ using $(MN)' = N'M'$: $(A^{-1})'A' = I' = I$. Transpose $A^{-1}A = I$: $A'(A^{-1})' = I$. So $(A^{-1})'$ is a two-sided inverse of A' . $\square$

Proposition 6.4 (The 2×2 formula). Slide statement. If $ad - bc \neq 0$, then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

If $ad - bc = 0$, no inverse exists.

Proof. Multiply:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = (ad - bc) I,$$

and the product in the other order gives the same matrix. Dividing by $ad - bc \neq 0$ gives I on both sides.

For the converse: if $ad - bc = 0$, the same computation shows $A \operatorname{adj}(A) = O$ where $\operatorname{adj}(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. If A had an inverse, multiplying on the left by A^{-1} would give $\operatorname{adj}(A) = O$, so $a = b = c = d = 0$ and $A = O$, which has no inverse either ($OB = O \neq I$). Contradiction. $\square$

7 Rules for Determinants

The slide “Rules for Determinants” lists eight rules. This section proves them at 2×2 in full, states the general definition, and says precisely what is being taken from a reference for larger sizes.

7.1 The 2×2 case, in full

Write $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det A = ad - bc$.

Proposition 7.1. *For 2×2 matrices:*

- (i) *Swapping the rows multiplies $\det$ by -1 .*
- (ii) *Scaling a row by c multiplies $\det$ by c .*
- (iii) *Adding a multiple of one row to the other leaves $\det$ unchanged.*
- (iv) *A zero row, or two equal rows, gives $\det = 0$.*
- (v) $\det(AB) = \det A \cdot \det B$.
- (vi) $\det(A') = \det A$.
- (vii) $\det I = 1$; if A is invertible, $\det(A^{-1}) = 1/\det A$.
- (viii) *If A is triangular, $\det A = ad$.*

Proof. (i) $\det \begin{pmatrix} c & d \\ a & b \end{pmatrix} = cb - da = -(ad - bc)$.

(ii) $\det \begin{pmatrix} ka & kb \\ c & d \end{pmatrix} = kad - kbc = k(ad - bc)$; the second row is the same by (i) applied twice.

(iii) $\det \begin{pmatrix} a & b \\ c+ka & d+kb \end{pmatrix} = a(d+kb) - b(c+ka) = ad + kab - bc - kab = ad - bc$.

(iv) A zero row: one of the two products in $ad - bc$ vanishes in each term, so $\det = 0$. Equal rows: $\det \begin{pmatrix} a & b \\ a & b \end{pmatrix} = ab - ba = 0$.

(v) With $B = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$, $AB = \begin{pmatrix} ap+br & aq+bs \\ cp+dr & cq+ds \end{pmatrix}$ and

$$\begin{aligned} \det(AB) &= (ap + br)(cq + ds) - (aq + bs)(cp + dr) \\ &= apcq + apds + brcq + brds - aqcp - aqdr - bscp - bsdr \\ &= apds + brcq - aqdr - bscp \\ &= ad(ps - qr) - bc(ps - qr) = (ad - bc)(ps - qr) = \det A \cdot \det B. \end{aligned}$$

(vi) $\det \begin{pmatrix} a & c \\ b & d \end{pmatrix} = ad - cb = ad - bc$.

(vii) $\det I = 1 \cdot 1 - 0 \cdot 0 = 1$. Then by (v), $1 = \det I = \det(AA^{-1}) = \det A \cdot \det(A^{-1})$.

(viii) Upper triangular means $c = 0$: $\det = ad - b \cdot 0 = ad$. Lower triangular means $b = 0$: same. □

7.2 The general case

Definition 7.2 (Determinant, cofactor expansion). For an $n \times n$ matrix A , let A_{1j} denote the $(n-1) \times (n-1)$ matrix obtained by deleting row 1 and column j . Define recursively

$$\det A = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det A_{1j},$$

with $\det(a) = a$ for 1×1 . For $n = 2$ this gives $ad - bc$; for $n = 3$ it is the expansion on the slide.

Theorem 7.3 (Rules (i)–(viii) hold for every n). *All eight rules of Proposition 7.1 hold for $n \times n$ matrices, with (viii) reading: the determinant of a triangular matrix is the product of its diagonal entries.*

What is proved here, and what is cited. Rule (viii) and rule (iv) for a zero row follow directly from Definition 7.2 by induction on n : expand along the first row. For an upper triangular matrix, only a_{11} in the first row can be nonzero, and A_{11} is again upper triangular, so $\det A = a_{11} \det A_{11} = a_{11} a_{22} \cdots a_{nn}$ by the induction hypothesis. For lower triangular, expand instead along the first column (which needs rule (vi) or a direct argument; see below). A zero row $i > 1$ propagates to every A_{1j} , which by induction has determinant 0; a zero first row makes every $a_{1j} = 0$.

Rules (i), (ii), (iii), (v), (vi) for general n require the fact that the determinant is the unique function of the rows that is linear in each row, alternating (sign flip under a swap), and equal to 1 on I . Proving that from Definition 7.2 is a page of bookkeeping that this course does not need. A complete proof is in Strang, *Introduction to Linear Algebra*, Chapter 5, or Lay, *Linear Algebra and Its Applications*, Sections 3.1–3.2. Rule (vii) follows from (v) exactly as in the 2×2 case once (v) is known. $\square$

Remark 7.4. The slides use, at 3×3 and above, only rules (iv), (viii), and the effect of the three row operations. Those are precisely what Section 8 below needs.

8 Pivots, Rank, and Invertibility

Proposition 8.1 (Triangular determinant). Slide statement. *If U is upper triangular, $\det U$ is the product of its diagonal entries (its pivots, when U is an echelon form).*

Proof. This is rule (viii), proved by induction in Theorem 7.3. $\square$

Theorem 8.2 (Determinant, pivots, and the inverse). Slide statement. *For an $n \times n$ matrix A , the following are equivalent:*

1. A^{-1} exists.
2. $\det A \neq 0$.
3. $\text{rank } A = n$.
4. The columns of A are linearly independent.
5. Elimination produces n pivots (no zero row).

6. $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $\mathbf{b} \in \mathbb{R}^n$.

Proof. Reduce A to an echelon form U by row operations. Since A is square, U is upper triangular.

(2) $\Leftrightarrow$ (5). Each row operation multiplies the determinant by a known nonzero factor: -1 for a swap, $c \neq 0$ for a scale, 1 for an add (rules (i)–(iii)). So $\det A = \kappa \det U$ with $\kappa \neq 0$, and by Proposition 8.1 $\det U$ is the product of the diagonal entries of U . That product is nonzero exactly when all n diagonal entries are nonzero, which is exactly when U has n pivots. (If some diagonal entry is 0, the staircase has skipped a column and the last row of U is zero.)

(5) $\Leftrightarrow$ (3). By definition, $\text{rank } A$ is the number of pivots.

(5) $\Leftrightarrow$ (4). The columns are dependent exactly when $A\mathbf{c} = \mathbf{0}$ has a nonzero solution. Row-reduce $[A \mid \mathbf{0}]$: the right column stays $\mathbf{0}$ throughout. If there are n pivots, back-substitution forces every $c_j = 0$, so only the trivial solution: independent. If fewer than n pivots, some column has no pivot; set that unknown to 1 and the other free unknowns to 0, back-substitute, and obtain a nonzero solution: dependent.

(5) $\Rightarrow$ (6). Row-reduce $[A \mid \mathbf{b}]$. With n pivots on the left, no row of the form $(0 \cdots 0 \mid c)$ can arise, and back-substitution determines every unknown uniquely. So there is exactly one solution, whatever $\mathbf{b}$ is.

(6) $\Rightarrow$ (1). By (6), for each standard basis vector $\mathbf{e}_j$ there is a unique $\mathbf{x}_j$ with $A\mathbf{x}_j = \mathbf{e}_j$. Let B be the matrix with columns $\mathbf{x}_1, \dots, \mathbf{x}_n$. Then AB has columns $A\mathbf{x}_j = \mathbf{e}_j$, so $AB = I$. To see that also $BA = I$: from $AB = I$, $A(BA) = (AB)A = A$, so $A(BA - I) = O$. Each column $\mathbf{w}$ of $BA - I$ satisfies $A\mathbf{w} = \mathbf{0}$; by (6) with $\mathbf{b} = \mathbf{0}$, the only solution is $\mathbf{w} = \mathbf{0}$ (since $\mathbf{0}$ is certainly one solution and it is unique). Hence $BA - I = O$.

(1) $\Rightarrow$ (5). Suppose A^{-1} exists but elimination gives fewer than n pivots. Then by the argument in (5) $\Leftrightarrow$ (4) there is $\mathbf{c} \neq \mathbf{0}$ with $A\mathbf{c} = \mathbf{0}$. Multiply by A^{-1} : $\mathbf{c} = A^{-1}\mathbf{0} = \mathbf{0}$. Contradiction.

The chain (1) $\Rightarrow$ (5) $\Rightarrow$ (6) $\Rightarrow$ (1) together with (5) $\Leftrightarrow$ (2),(3),(4) covers all six. $\square$

Proposition 8.3 (Rank criterion for $A\mathbf{x} = \mathbf{b}$). Slide statement. Let A be $m \times n$.

- If $\text{rank } A \neq \text{rank}[A \mid \mathbf{b}]$: no solution.
- If $\text{rank } A = \text{rank}[A \mid \mathbf{b}] = n$: exactly one solution.
- If $\text{rank } A = \text{rank}[A \mid \mathbf{b}] < n$: infinitely many solutions.

Proof. Row-reduce $[A \mid \mathbf{b}]$ to echelon form $[U \mid \mathbf{d}]$. The same operations reduce A to U , so $\text{rank } A$ is the number of pivots in U , and $\text{rank}[A \mid \mathbf{b}]$ is the number of pivots in $[U \mid \mathbf{d}]$. The only way the augmented matrix can have *more* pivots than U is a row of the form $(0 \cdots 0 \mid d_i)$ with $d_i \neq 0$, i.e. a pivot in the last column; it can never have fewer.

Case $\text{rank } A \neq \text{rank}[A \mid \mathbf{b}]$. Then such a row exists. It reads $0 = d_i \neq 0$, which no $\mathbf{x}$ satisfies. No solution.

Case *ranks equal*. No such row, so every equation with a zero left side has a zero right side and is vacuous; the system is consistent. The $n - \text{rank } A$ columns without a pivot correspond to free unknowns. If $\text{rank } A = n$, there are none, and back-substitution gives one solution. If $\text{rank } A < n$, there is at least one free unknown; each choice of its value yields a distinct solution, so there are infinitely many. $\square$

9 Regression in Matrix Form

Throughout, X is $n \times p$ with $p = k + 1$ (a column of ones plus k predictors), $\mathbf{y} \in \mathbb{R}^n$, and $\text{SSR}(\boldsymbol{\beta}) = (\mathbf{y} - X\boldsymbol{\beta})'(\mathbf{y} - X\boldsymbol{\beta})$.

Lemma 9.1 (Gradient of a linear form). Slide statement. For fixed $\mathbf{c} \in \mathbb{R}^p$, $\nabla_{\boldsymbol{\beta}}(\mathbf{c}'\boldsymbol{\beta}) = \mathbf{c}$.

Proof. $\mathbf{c}'\boldsymbol{\beta} = \sum_{j=1}^p c_j \beta_j$. The partial derivative with respect to β_j is c_j . Stack them: the gradient is $\mathbf{c}$. $\square$

Lemma 9.2 (Gradient of a quadratic form). Slide statement. For a symmetric $p \times p$ matrix M , $\nabla_{\boldsymbol{\beta}}(\boldsymbol{\beta}'M\boldsymbol{\beta}) = 2M\boldsymbol{\beta}$.

Proof. $\boldsymbol{\beta}'M\boldsymbol{\beta} = \sum_{i=1}^p \sum_{l=1}^p m_{il} \beta_i \beta_l$. Differentiate with respect to β_j . The variable β_j appears in the terms with $i = j$ (giving $\sum_l m_{jl} \beta_l$), in the terms with $l = j$ (giving $\sum_i m_{ij} \beta_i$), and the term $i = l = j$ is counted in both, correctly, since $\partial(m_{jj} \beta_j^2) / \partial \beta_j = 2m_{jj} \beta_j$. So

$$\frac{\partial}{\partial \beta_j}(\boldsymbol{\beta}'M\boldsymbol{\beta}) = \sum_l m_{jl} \beta_l + \sum_i m_{ij} \beta_i = \sum_l m_{jl} \beta_l + \sum_i m_{ji} \beta_i = 2 \sum_l m_{jl} \beta_l = 2(M\boldsymbol{\beta})_j,$$

where symmetry $m_{ij} = m_{ji}$ was used in the second equality. $\square$

Proposition 9.3 (The normal equations). Slide statement. $\nabla \text{SSR}(\boldsymbol{\beta}) = -2X'\mathbf{y} + 2X'X\boldsymbol{\beta}$, so the FOC is $X'X\boldsymbol{\beta} = X'\mathbf{y}$.

Proof. Expand, using $(X\boldsymbol{\beta})' = \boldsymbol{\beta}'X'$ and the fact that a 1×1 matrix equals its transpose (so $\mathbf{y}'X\boldsymbol{\beta} = \boldsymbol{\beta}'X'\mathbf{y}$):

$$\text{SSR}(\boldsymbol{\beta}) = \mathbf{y}'\mathbf{y} - \mathbf{y}'X\boldsymbol{\beta} - \boldsymbol{\beta}'X'\mathbf{y} + \boldsymbol{\beta}'X'X\boldsymbol{\beta} = \mathbf{y}'\mathbf{y} - 2(X'\mathbf{y})'\boldsymbol{\beta} + \boldsymbol{\beta}'(X'X)\boldsymbol{\beta}.$$

The first term does not depend on $\boldsymbol{\beta}$. Apply Lemma 9.1 with $\mathbf{c} = X'\mathbf{y}$ to the second and Lemma 9.2 with $M = X'X$ (symmetric, since $(X'X)' = X'X$) to the third: $\nabla \text{SSR} = -2X'\mathbf{y} + 2X'X\boldsymbol{\beta}$. Set to $\mathbf{0}$ and divide by 2. $\square$

Theorem 9.4 (The rank condition). Slide statement. $X'X$ is invertible if and only if the columns of X are linearly independent, i.e. $\text{rank } X = p$.

Proof. $X'X$ is $p \times p$. By Theorem 8.2, it is invertible if and only if its columns are independent, i.e. if and only if $X'X\mathbf{c} = \mathbf{0}$ has only the solution $\mathbf{c} = \mathbf{0}$. So it suffices to show: $X'X\mathbf{c} = \mathbf{0} \iff X\mathbf{c} = \mathbf{0}$.

($\Leftarrow$) If $X\mathbf{c} = \mathbf{0}$, multiply by X' .

($\Rightarrow$) If $X'X\mathbf{c} = \mathbf{0}$, multiply on the left by $\mathbf{c}'$: $\mathbf{c}'X'X\mathbf{c} = 0$, i.e. $(X\mathbf{c})'(X\mathbf{c}) = \|X\mathbf{c}\|^2 = 0$. A vector of length zero is the zero vector, so $X\mathbf{c} = \mathbf{0}$.

Hence $X'X\mathbf{c} = \mathbf{0}$ has only the trivial solution exactly when $X\mathbf{c} = \mathbf{0}$ does, which is exactly when the columns of X are independent. $\square$

Remark 9.5 (When it fails). If $\text{rank } X < p$ the normal equations are still consistent (they are the FOC of a convex function that is bounded below, so a minimizer exists), but by Proposition 8.3 they have infinitely many solutions: many $\boldsymbol{\beta}$ give the same fitted values $X\boldsymbol{\beta}$ and the same minimal SSR. The data cannot distinguish them. That is what “perfect multicollinearity” means.

Proposition 9.6 (The FOC is a minimum). Slide statement. Any $\hat{\beta}$ solving $X'X\beta = X'y$ is a global minimizer of SSR. When $X'X$ is invertible it is the unique minimizer, $\hat{\beta} = (X'X)^{-1}X'y$.

Proof. SSR is convex by Proposition 1.5 and has continuous partials (it is a polynomial in β). A solution of the normal equations is a point where $\nabla \text{SSR} = \mathbf{0}$ (Proposition 9.3), so by Theorem 1.6 it is a global minimum. If $X'X$ is invertible, the normal equations have exactly one solution (Theorem 8.2, item 6), and multiplying $X'X\beta = X'y$ on the left by $(X'X)^{-1}$ isolates it. $\square$

Example 9.7 (The ten states in matrix form). X is 10×2 with first column $\mathbf{1}$ and second column $\mathbf{x} = (-2, -1, -1, 0, 1, 1, 2, 3, 3, 4)'$. Then

$$X'X = \begin{pmatrix} \mathbf{1} \cdot \mathbf{1} & \mathbf{1} \cdot \mathbf{x} \\ \mathbf{x} \cdot \mathbf{1} & \mathbf{x} \cdot \mathbf{x} \end{pmatrix} = \begin{pmatrix} 10 & 10 \\ 10 & 46 \end{pmatrix}, \quad X'y = \begin{pmatrix} \mathbf{1} \cdot \mathbf{y} \\ \mathbf{x} \cdot \mathbf{y} \end{pmatrix} = \begin{pmatrix} 500 \\ 428 \end{pmatrix},$$

which are the two normal equations of Example 2.5. $\det(X'X) = 460 - 100 = 360 \neq 0$, so by Theorem 9.4 the columns $\mathbf{1}$ and $\mathbf{x}$ are independent (relative unemployment is not constant), and

$$\hat{\beta} = \frac{1}{360} \begin{pmatrix} 46 & -10 \\ -10 & 10 \end{pmatrix} \begin{pmatrix} 500 \\ 428 \end{pmatrix} = \frac{1}{360} \begin{pmatrix} 18720 \\ -720 \end{pmatrix} = \begin{pmatrix} 52 \\ -2 \end{pmatrix}.$$